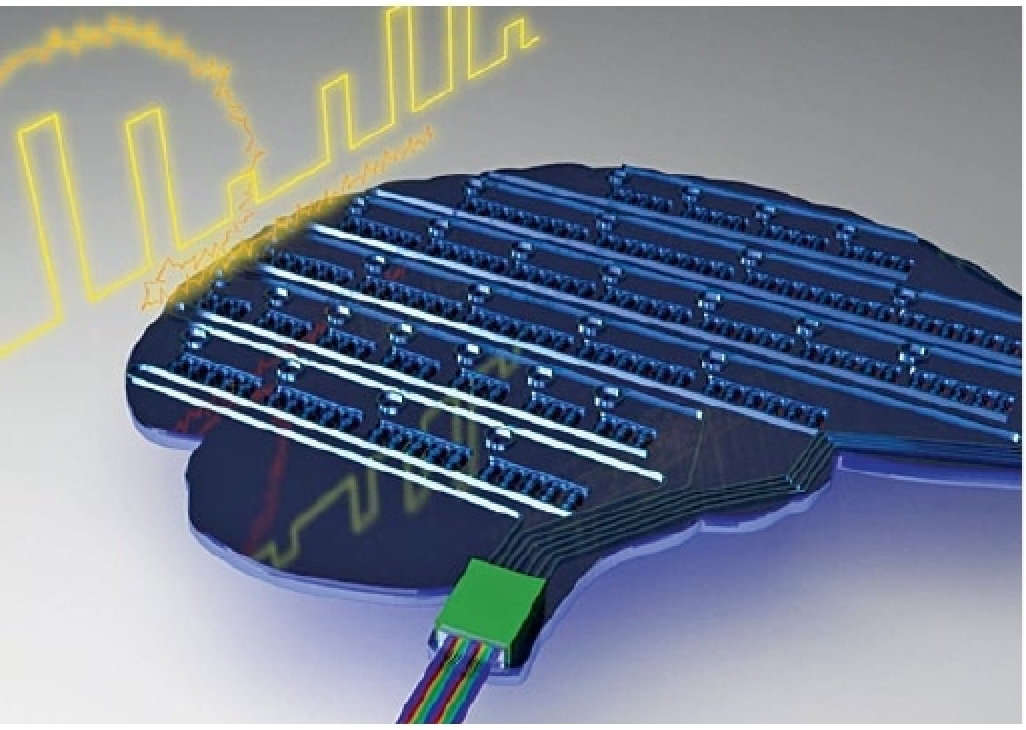
A step towards light-based brain-like computing chip

A team of European scientists from Germany and the UK has revealed a pioneering new way to create optical neural networks that teach themselves to recognise patterns in image data.

Traditional computers are built on von Neumann architecture, with separate memory and processor units operating one command at a time. Compared to the brain—where processing and memory functions are co-located and a massively parallel approach is used—this can be very inefficient.

To develop computers that work more like the brain, hardware devices that operate in a similar way to brain neurons and synapses are needed, and such devices should be combined into large-scale networks capable of real-world tasks.



Representative image (Credit: University of Oxford)

Prof. Wolfram Pernice from University of Muenster, lead partner in the study, explains, "We have made significant steps towards this goal—working here with light-based devices rather than electronics—demonstrating integrated photonic neurosynaptic networks that can recognise patterns, identify letters and numbers, even correctly differentiate between the languages of written text."

Prof. Harish Bhaskaran, co-author from Oxford University Department of Materials, adds, "Working with photons instead of electrons will allow us to exploit well-known benefits of optical technologies—wavelength division multiplexing, ultra-high bandwidths and low energy consumption—but here in the realm of computing rather than the more usual communications field."

University of Muenster's Johannes Feldmann points out, "Key to our work is the successful merging of phase-change devices and silicon photonics. This gives us the ability to successfully mimic the behaviour of biological neurons and synapses, at least in a basic way."

Prof. C David Wright, co-author of the study from University of Exeter, sums up, "This is a significant experimental milestone—a fully-scalable integrated photonic system that can process and store information in a brain-like fashion. Our approach could find widespread utility in power-critical situations such as mobile and so-called Edge computing applications."

Ai-Da, the first ultra-realistic humanoid AI robot artist

Named after Ada Lovelace, the first female computer programmer, Ai-Da the robot is the brainchild of gallery director Aidan Meller. She has been designed and built by Cornish robotics company Engineered Arts, and her drawing abilities have been created and developed by students at University of Leeds. Ai-Da is the first ultra-realistic humanoid artificial intelligence (AI) robot artist in the world.

Ai-Da is a mechanical robot—she is not real and has no thoughts and feelings—but she foretells a period when transhuman biotechnology could be possible. Through the vehicle of an ultra-realistic robot, Ai-Da, the creators are looking at the boundaries of AI, technology and organic life, and how humans interact with it.

Ai-Da, the world's first robot artist, with some of her work (Credit: metro.co.uk)